EXERCISE 10.4

Find the derivatives of the following (1 – 18) :

(1) $y = x^{\cos x}$

(2) $y = x^{\log x} + (\log x)^x$

(3) $\sqrt{xy} = e^{(x-y)}$

(4) $x^y = y^x$

(5) $(\cos x)^{\log x}$

(6) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(7) $\sqrt{x^2 + y^2} = \tan^{-1}\left(\frac{y}{x}\right)$

(8) $\tan(x + y) + \tan(x - y) = x$

(9) If $\cos(xy) = x$, show that $\frac{dy}{dx} = \frac{-(1 + y\sin(xy))}{x\sin xy}$

(10) $\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}$

(11) $\tan^{-1}\left(\frac{6x}{1-9x^2}\right)$

(12) $\cos\left(2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right)$

(13) $x = a\cos^3 t$; $y = a\sin^3 t$

(14) $x = a(\cos t + t\sin t)$; $y = a(\sin t - t\cos t)$

(15) $x = \frac{1-t^2}{1+t^2}, y = \frac{2t}{1+t^2}$

(16) $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

(17) $\sin^{-1}(3x - 4x^3)$

(18) $\tan^{-1}\left(\frac{\cos x + \sin x}{\cos x - \sin x}\right)$.

(19) Find the derivative of $\sin x^2$ with respect to x^2.

(20) Find the derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to $\tan^{-1} x$.

(21) If $u = \tan^{-1}\frac{\sqrt{1+x^2} - 1}{x}$ and $v = \tan^{-1} x$, find $\frac{du}{dv}$.

(22) Find the derivative with $\tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$ with respect to $\tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$.

(23) If $y = \sin^{-1} x$ then find y''.

(24) If $y = e^{\tan^{-1} x}$, show that $(1+x^2)y'' + (2x-1)y' = 0$.

(25) If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, show that $(1-x^2)y_2 - 3xy_1 - y = 0$.

(26) If $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ then prove that at $\theta = \frac{\pi}{2}$, $y'' = \frac{1}{a}$.

(27) If $\sin y = x\sin(a + y)$, then prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$, $a \neq n\pi$.

(28) If $y = (\cos^{-1} x)^2$, prove that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 2 = 0$. Hence find y_2 when $x = 0$

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